Bivariate Relationships

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Relationships Between Variables

Methods in Psychological Research

Bivariate

- So far, we have been analyzing summary statistics that describe aspects of a single list of numbers
- Frequently, however, we are interested in how variables behave together
    - Measuring the relationship (correlation)
    - Modeling the relationship (regression)

Motivating Example: Smoking and Lung Capacity

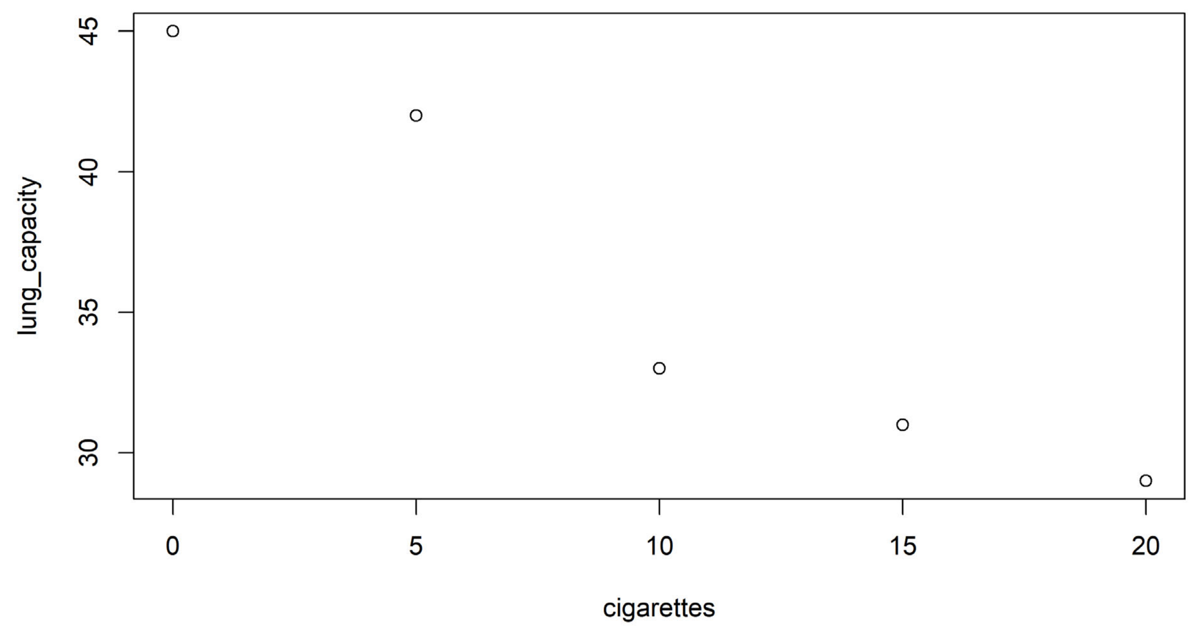
- Suppose, for example, we wanted to investigate the relationship between cigarette smoking and lung capacity
- We might ask a group of people about their smoking habits, and measure their lung capacities

cigarettes	lung_capacity
0	45
5	42
10	33
15	31
20	29

Visualizing

- With R or any other statistics software, we can produce a scatterplot, like this one.

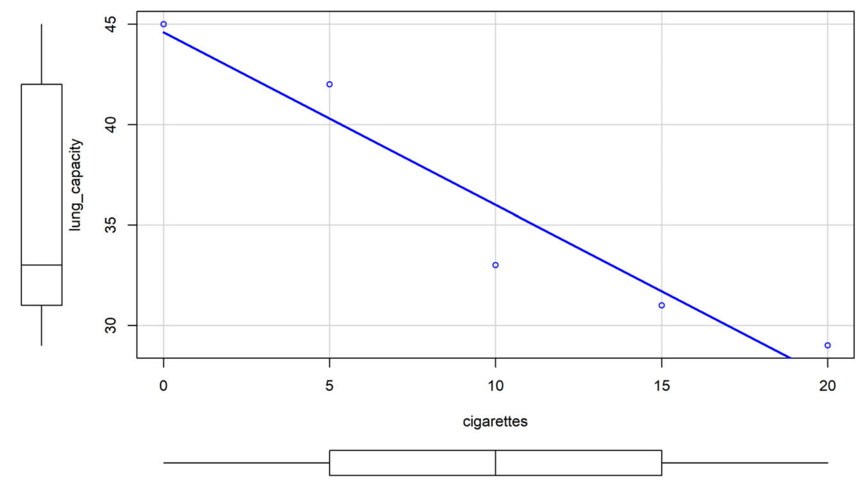
```
plot(smoking)
```



Or this one

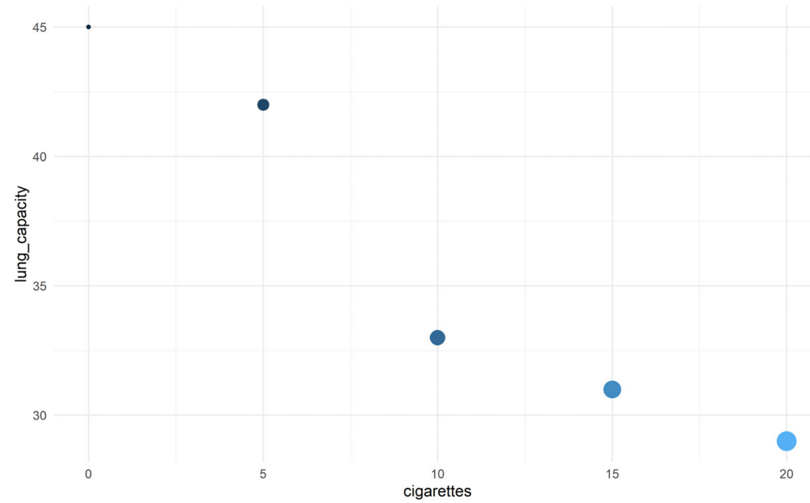
- A scatterplot shows the relationship between two quantitative variables that are measured on the same individuals.
  - The values of one variable appear on the horizontal axis,
  - and the values of the other variable appear on the vertical axis.
  - Each individual in the data appears as the point in the plot fixed by the values of both variables for that individual.

```
g_capacity~cigarettes, data=smoking)
```



Or this one

- Always plot the explanatory variable,
  - if there is one, on the horizontal axis (the x-axis) of a scatterplot
- If there is no explanatory-response distinction,
  - either variable can go on the horizontal axis.




```r
library(ggplot2)
library(ggExtra)

# classic plot
p <- ggplot(smoking, aes(x=cigarettes,
                         y=lung_capacity,
                         color = cigarettes,
                         size=cigarettes)) +
      geom_point() + theme_minimal() +
      theme(legend.position="none")
p
```

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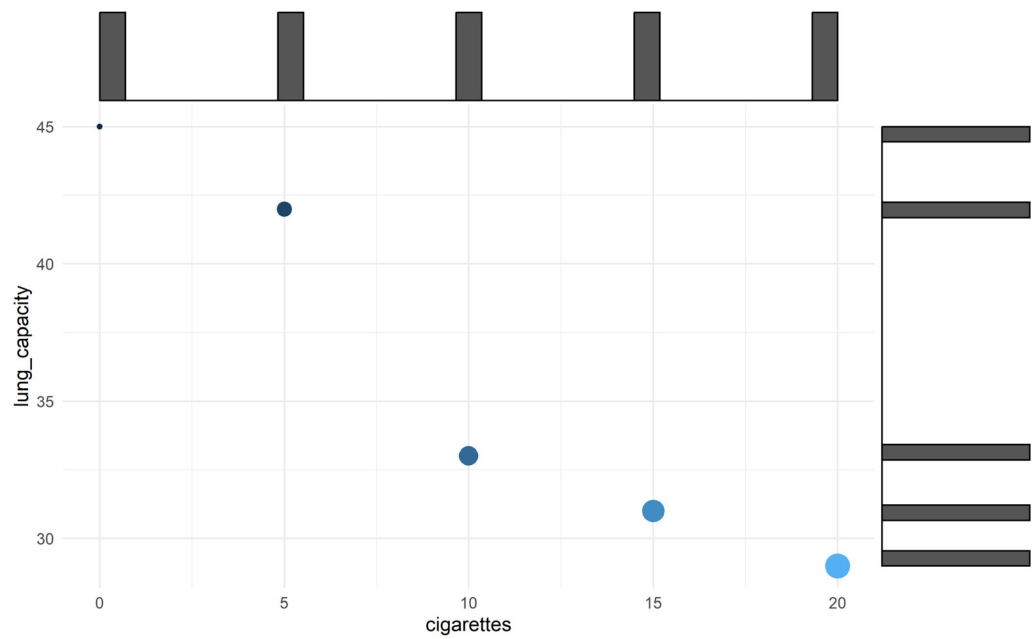
Or this one

- We can see from the graph that as smoking goes up, lung capacity tends to go down.

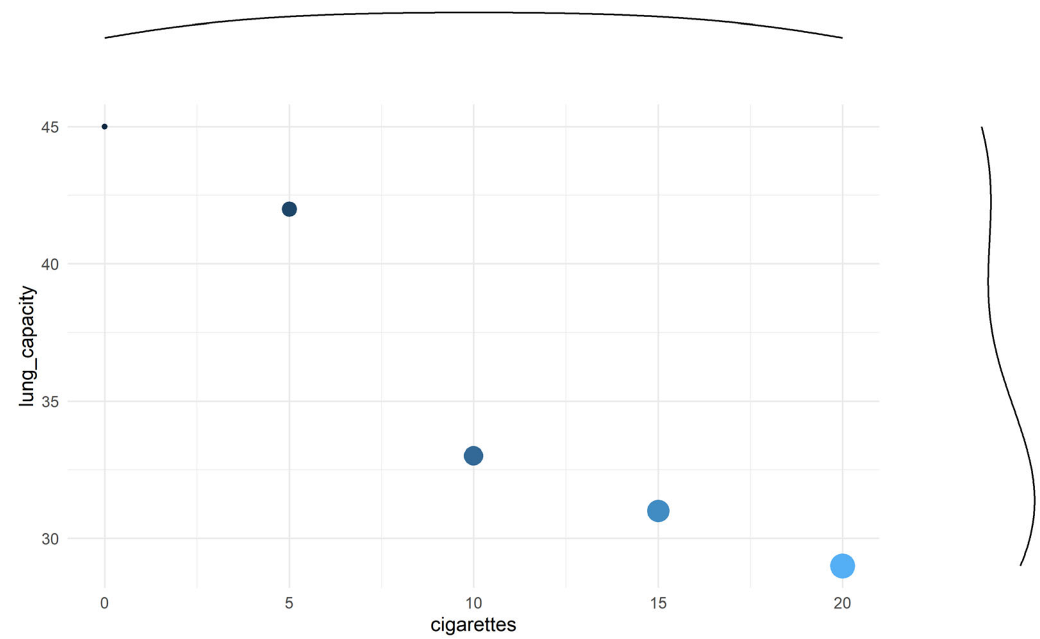
```
# with marginal histogram
ggMarginal(p, type ="histogram")
```



Or this one

- Here, the two variables covary in opposite directions.
- This is a negative relationship

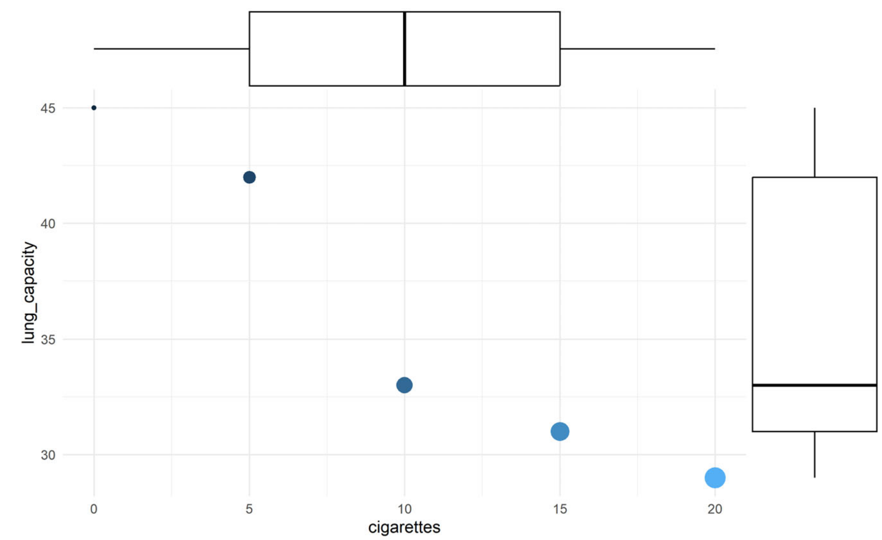
```r
# marginal density
ggMarginal(p, type="density")
```



Scatter Plots

- In any graph of data, look for the overall pattern and for striking deviations from that pattern.
- You can describe the overall pattern of a scatterplot by the
  - direction (positive or negative),
  - form, and
  - strength of the relationship.

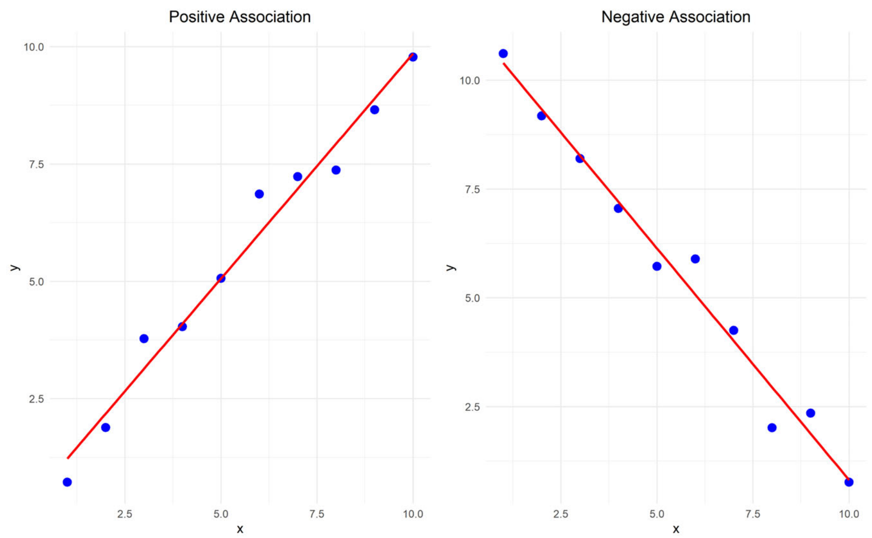
```r
# marginal boxplot
ggMarginal(p, type="boxplot")
```



- More examples `https://www.statmethods.net/graphs/scatterplot.html`

Direction of Association

- Two variables are **positively associated** when above-average values of one tend to accompany above-average values of the other, and below-average values also tend to occur together.
- Two variables are **negatively associated** when above-average values of one tend to accompany below-average values of the other, and vice versa.



Strength of the Relationship

- We now examine two statistics for quantifying how variables covary:
  - covariance and
  - correlation.

Covariance

Covariance

- Covariance is a measure of how much two random variables vary together.
- When two variables covary in opposite directions, as smoking and lung capacity do,
  - values tend to be on opposite sides of the group mean.
  - That is, when smoking is above its group mean,
  - lung capacity tends to be below its group mean.
- Consequently, by averaging the product of deviation scores, we can obtain a measure of how the variables vary together.

Sample Covariance

- Sample covariance between X and Y is an estimate of that average (cross-) product of deviation scores in the population

$$s_{x,y} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

- A more computationally convenient formula:

$$s_{x,y} = \frac{1}{n-1}\left(\sum_{i=1}^{n} x_i y_i - \frac{\sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n}\right)$$

- Useful fact: the variance of a variable is its covariance with itself.

$$s^2 = s_{x,x} = \frac{1}{n-1}\sum_{i=1}^{n}((x_i - \bar{x})(x_i - \bar{x}))$$

Computing Covariance

- If we wanted to compute the covariance for smoking and lung capacity, we can save ourselves some heartache.
- Rather than finding each deviation from the mean for x and y

```
covariance_calc=data.frame(X=smoking$cigarettes,Y=smoking$lung_capacity)

# find x_i-x_bar
covariance_calc$dx=covariance_calc$X-mean(covariance_calc$X)

# find y_i-y_bar
covariance_calc$dy=covariance_calc$Y-mean(covariance_calc$Y)
```

- and then finding their product

```
#cross multiply x deviations and y deviations
covariance_calc$dxdy=covariance_calc$dy*covariance_calc$dx

# cross multi X and Y
covariance_calc$xy=covariance_calc$Y*covariance_calc$X
```

Computing Covariance

- Giving these values

```
kable(covariance_calc)
```

X	Y	dx	dy	dxdy	xy
0	45	-10	9	-90	0
5	42	-5	6	-30	210
10	33	0	-3	0	330
15	31	5	-5	-25	465
20	29	10	-7	-70	580

- We'd then sum the dXdY column and get -215, and we then compute the covariance as:

$$s_{x,y} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) = \frac{-215}{5-1} = -53.75$$

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Alternative Computation

- We can compute $\sum x = 50, \sum y = 180, \sum xy = 1585$, and n, and use that easier equation

$$
\begin{aligned}
s_{x,y} &= \frac{1}{n-1}\left(\sum x_i y_i - \frac{\sum x_i \sum y_i}{n}\right) \\
&= \frac{1}{5-1}\left(1585 - \frac{50 \times 180}{5}\right) \\
&= \frac{1}{4}\left(1585 - \frac{9000}{5}\right) \\
&= \frac{1}{4}(1585 - 1800) \\
&= \frac{1}{4}(-215) \\
&= -53.75
\end{aligned}
$$

Covariance in R

```
cov(smoking)
```

```
##                  cigarettes lung_capacity
## cigarettes            62.50        -53.75
## lung_capacity        -53.75         50.00
```

```
cov(smoking)[1,2]
```

```
## [1] -53.75
```

```
var(smoking$cigarettes)
```

```
## [1] 62.5
```

Covariance

- Although covariance is an *extremely* important concept and is the cornerstone of many advanced methods (ANOVA, ANCOVA, SEM, regression, etc), it has some limitations:
- it has interpretation problems just like variance
- it isn't in a meaningful scale
- it tells us whether a relationship is positive or negative
  - but not much more than that.

Covariance Scaling

- Is -53.75 a strong relationship?
  - It depends on the scale
- If we convert cigarettes into packs, the relationship hasn't changed.
- but the covariance has...

```
smoking$packs=smoking$cigarettes/20

cov(smoking$packs,smoking$lung_capacity)
```

```
## [1] -2.6875
```

- The value is, in a sense, "polluted by the metric of the numbers."
- Depending on the scale of the data, the absolute value of the covariance can be very large or very small

Standardizing Covariance

- We can take the scale out of the covariance.
- What happens if we use z-scores instead of raw deviations?
- Remember, that z-scores are also a measure of deviations
- This is called the (Pearson) Correlation Coefficient

Correlation Coefficient

- The sample correlation is the sum of cross-products of z-scores divided by n-1:

$$r_{x,y} = \frac{1}{n-1}\sum_{i=1}^{n}(Z_{x_i} \times Z_{y_i})$$

- The sample correlation is the sum of cross-products of z-scores divided by n:

$$\rho_{x,y} = \frac{1}{n}\sum_{i=1}^{n}(Z_{x_i} \times Z_{y_i})$$

Formulae for Correlation

- We can think of a correlation coefficient as a covariance with the standard deviations factored out:

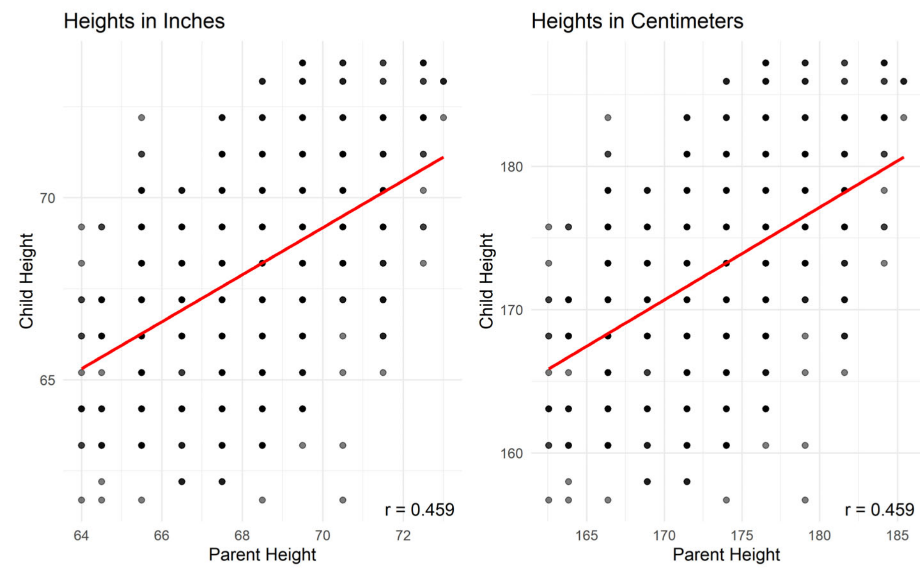
$$r_{x,y} = \frac{s_{x,y}}{s_x \times s_y}$$

- we can think of covariance as a correlation with the standard deviations put back in.

$$s_{x,y} = r_{x,y} s_x s_y$$

Properties of r

- Notation
  - Sample Statistic r
  - Population Parameter ρ
    - pronounced row, but spelt rho
- Transformation
  - r is the same regardless of how the values are measured.
  - Height in inches and centimeters lead to the same r



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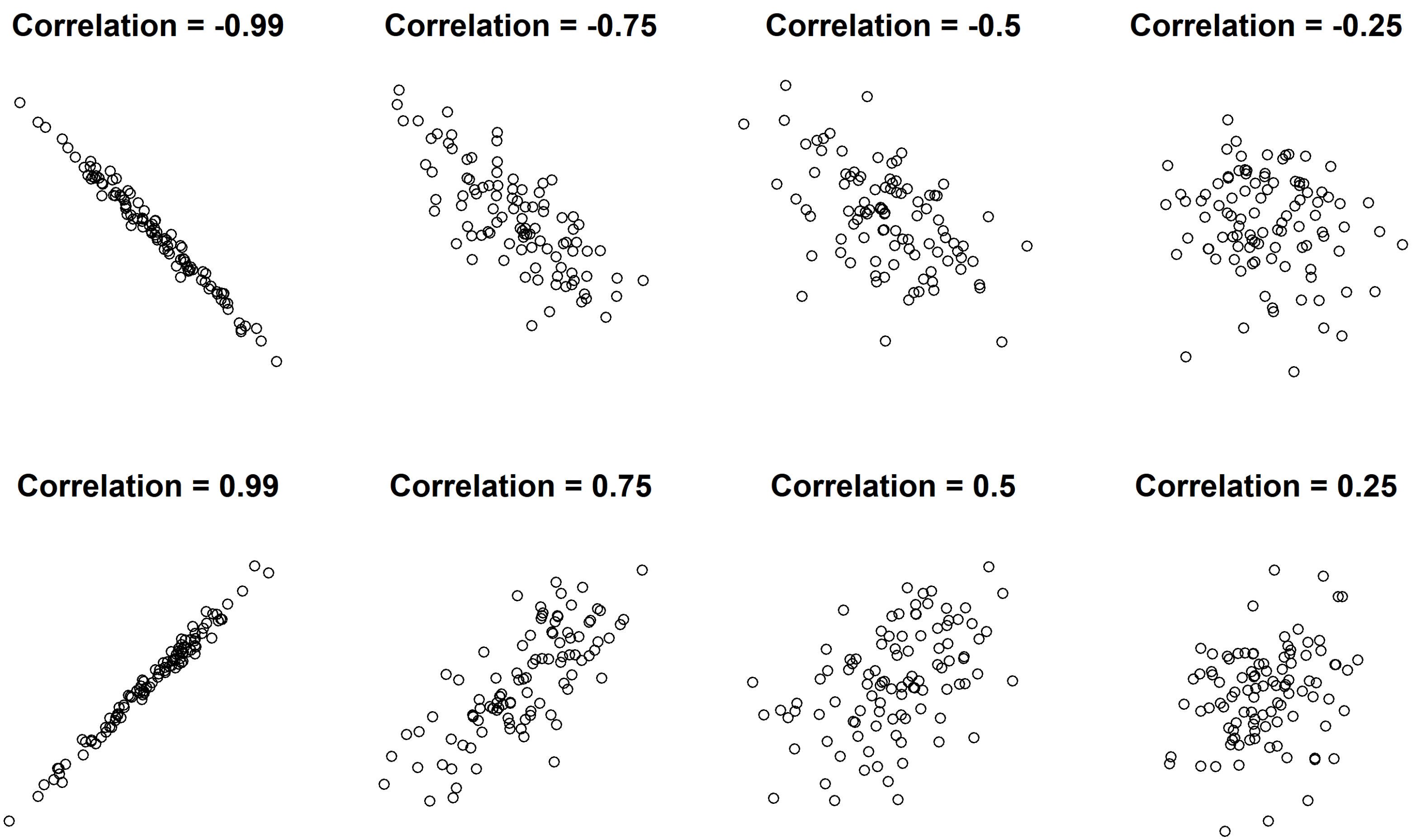
Properties of r

- r ranges from -1 to 1 [-1,1]
  - |r| indicates the size of the relationship
  - r>0 indicates positive relationship (0,1]
  - r<0 indicates negative relationship [-1,0)
  - r is symmetric
    - $r_{x,y} = r_{y,x}$

Correlation Magnitude



Correlation Interpretation

- Perfect
  - Exactly –1. A perfect downhill (negative) linear relationship
  - Exactly +1. A perfect uphill (positive) linear relationship
- Strong
  - -0.70. A strong downhill (negative) linear relationship
  - +0.70. A strong uphill (positive) linear relationship
- Moderate
  - –0.50. A moderate downhill (negative) relationship
  - +0.50. A moderate uphill (positive) relationship
- Small
  - –0.30. A small downhill (negative) linear relationship
  - +0.30. A small uphill (positive) linear relationship

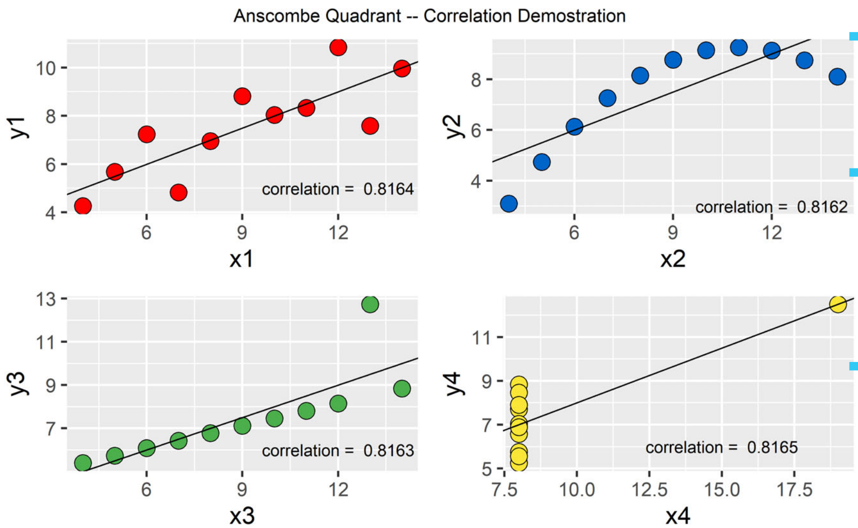
- Develop your correlation intuition `http://guessthecorr`

Linear Relationship

+ The correlation quantifies the linear association.
+ If your data aren't linear, then the correlation can be misleading.
+ If your data have outliers, then the correlation can be misleading.

Anscombe's Quartet

- Anscombe's Quartet comprises four datasets that have nearly identical simple descriptive statistics, yet appear very different when graphed.



- These data were constructed in 1973 by the statistician Francis Anscombe (1973).
- Demonstrates the importance of graphing data before analyzing it and the effect of outliers on statistical properties.
- It's intended to counter the impression among statisticians that "numerical calculations are exact, but graphs are rough."

Correlation Cautions

- Correlation requires that both variables be quantitative, so it makes sense to do the arithmetic indicated by the formula for r.
- Correlation does not describe curved relationships between variables, no matter how strong the relationship is between them.
- Correlation is strongly affected by a few outlying observations.
- Correlation is not a complete summary of two-variable data.

More Correlation Cautions

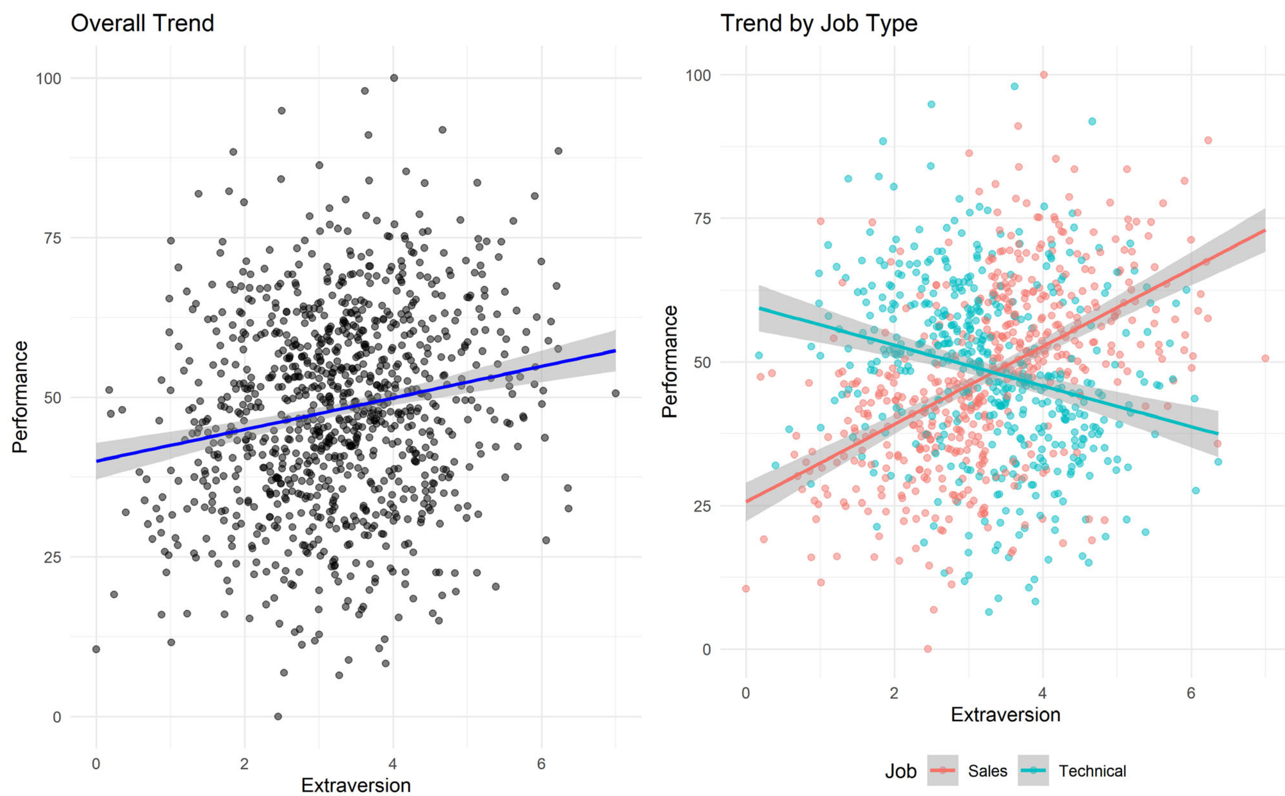
- Restriction of Range.
- Results in r to go down artificially
  - r -> ρ
- Combined groups can hid the relationships, when groups have different relationships
  - Simpson’s Paradox

Simpson's Paradox

Simpson's Paradox occurs when a trend appears in different groups of data but disappears or reverses when these groups are combined.
Let's explore this through two examples:

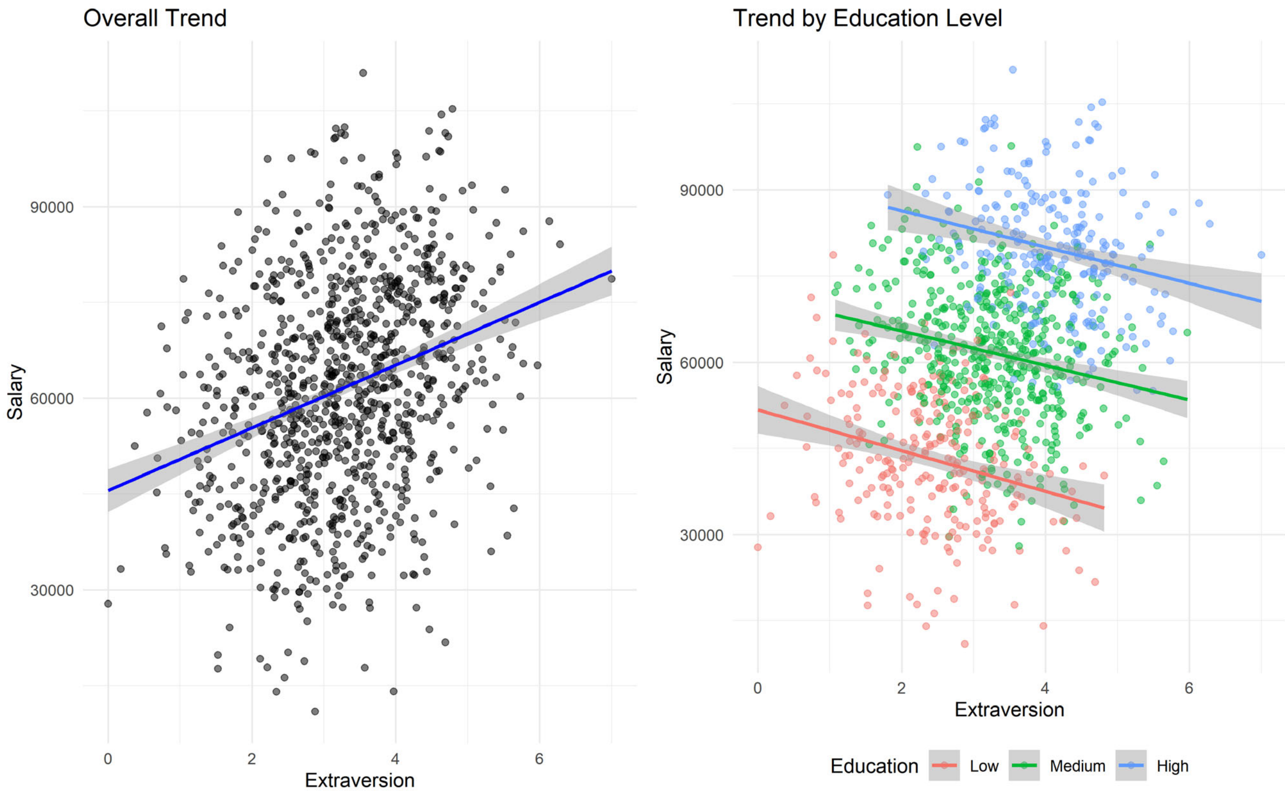
1. Extraversion and Job Performance
2. Extraversion, Education, and Salary

Example 1: Extraversion and Job Performance



- Left: Overall, there's a slight positive relationship between Extraversion and Performance.
- Right: When separated by job type, we see negative relationships within each group.
- This illustrates Simpson's Paradox: the trend in the overall data is different from the trends in the subgroups.

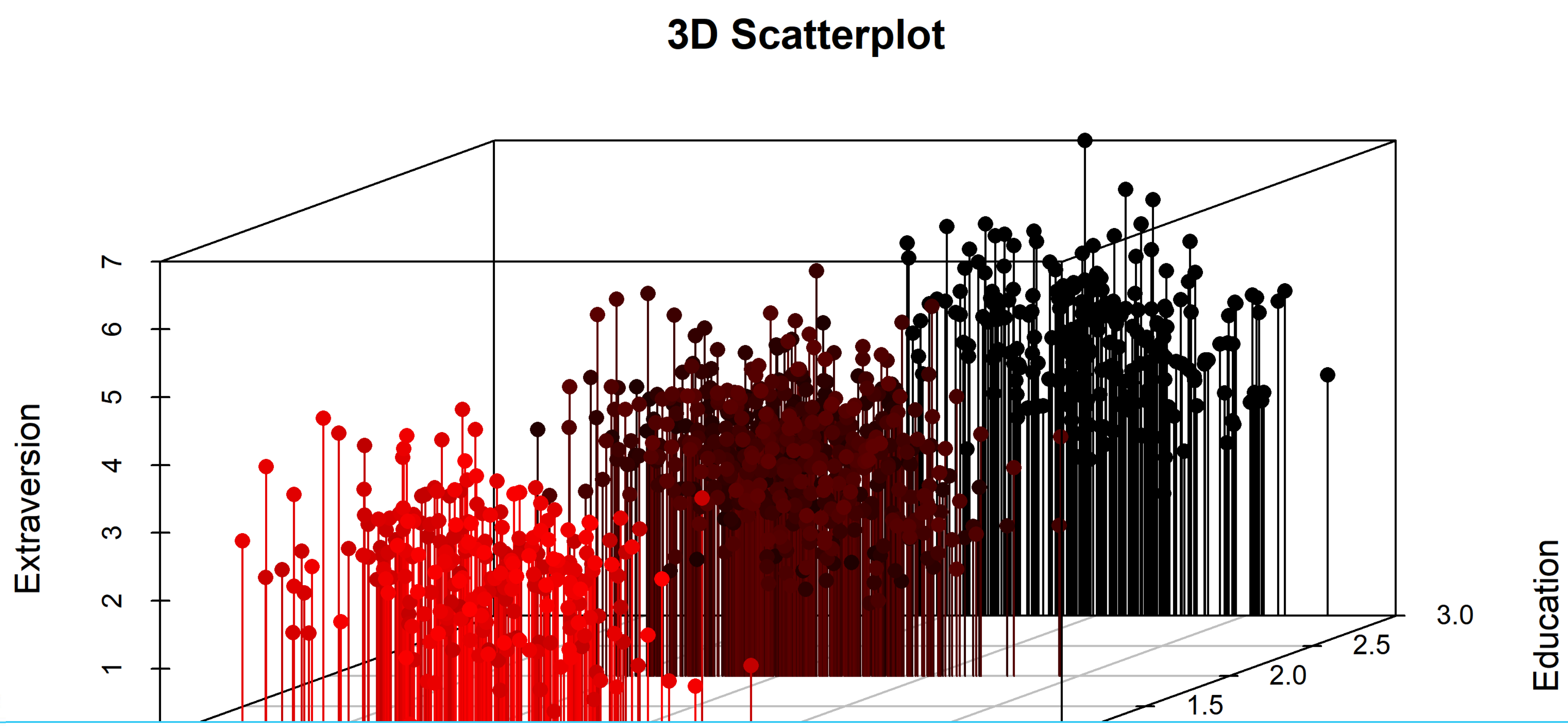
Example 2: Extraversion, Education, and Salary



- Left: Overall, there's a positive relationship between Extraversion and Salary.
- Right: When separated by Education level, we see negative relationships within each group.
- This is another example of Simpson's Paradox: the overall trend is reversed when we look at subgroups.

Simposon's Paradox

Illustrated



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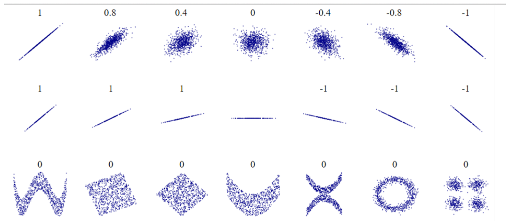
Final Comments



- The correlation coefficient is neither robust nor resistant.
  - Not robust because strong nonlinear relationships between the two variables may not be recognized.
  - Not resistant because it is sensitive to outlying points.
  - It is the most sensitive summary stat we have seen thus far

Shared Variance: R^2

- How much do those correlated variables have in common? `http://rpsychologist.com/d3/correlation/`
- Common variance
  - i.e. Shared Variance
- Coefficient of Determination
  - % variability between the two variables that has been accounted for
  - the remaining $1 - R^2$ of the variability is still unaccounted for

References

Magnitude Code

```
# Load necessary libraries
library(ggplot2)
library(MASS)
library(gridExtra)

# Function to create correlated data and plot
create_plot <- function(correlation) {
  Sigma <- matrix(c(1, correlation, correlation, 1), 2, 2)
  data <- as.data.frame(mvrnorm(n = 100, mu = c(0, 0), Sigma = Sigma))
  colnames(data) <- c("x", "y")
  plot <- ggplot(data, aes(x = x, y = y)) +
    geom_point(shape = 1) +
    ggtitle(paste("Correlation =", correlation)) +
    theme_minimal() +
    theme(
      plot.title = element_text(size = 12, hjust = 0.5, face = "bold"),
      axis.text = element_blank(),
      axis.ticks = element_blank(),
      panel.grid = element_blank(),
      axis.title = element_blank()
    ) +
    xlim(-3, 3) + ylim(-3, 3) +
    coord_fixed(ratio = 1)
  return(plot)
}


# List of correlations
correlations <- c(-0.99, -0.75, -0.5, -0.25,0.99, 0.75, 0.5, 0.25)

# Create list of plots
plots <- lapply(correlations, create_plot)

# Arrange plots in a grid
grid.arrange(grobs = plots, ncol = 4)
```

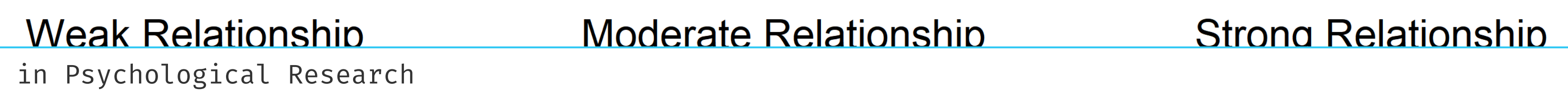
```
##
## scale
```

association code

```
library(ggplot2)
library(gridExtra)

# Create data for positive association
set.seed(123)  # for reproducibility
pos_data <- data.frame(
  x = 1:10,
  y = 1:10 + rnorm(10, sd = 0.5)
)

# Create data for negative association
neg_data <- data.frame(
  x = 1:10,
  y = 10:1 + rnorm(10, sd = 0.5)
)

# Plot for positive association
pos_plot <- ggplot(pos_data, aes(x, y)) +
  geom_point(color = "blue", size = 3) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  ggtitle("Positive Association") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5))

# Plot for negative association
neg_plot <- ggplot(neg_data, aes(x, y)) +
  geom_point(color = "blue", size = 3) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  ggtitle("Negative Association") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5))

# Combine plots
grid.arrange(pos_plot, neg_plot, ncol = 2)
```

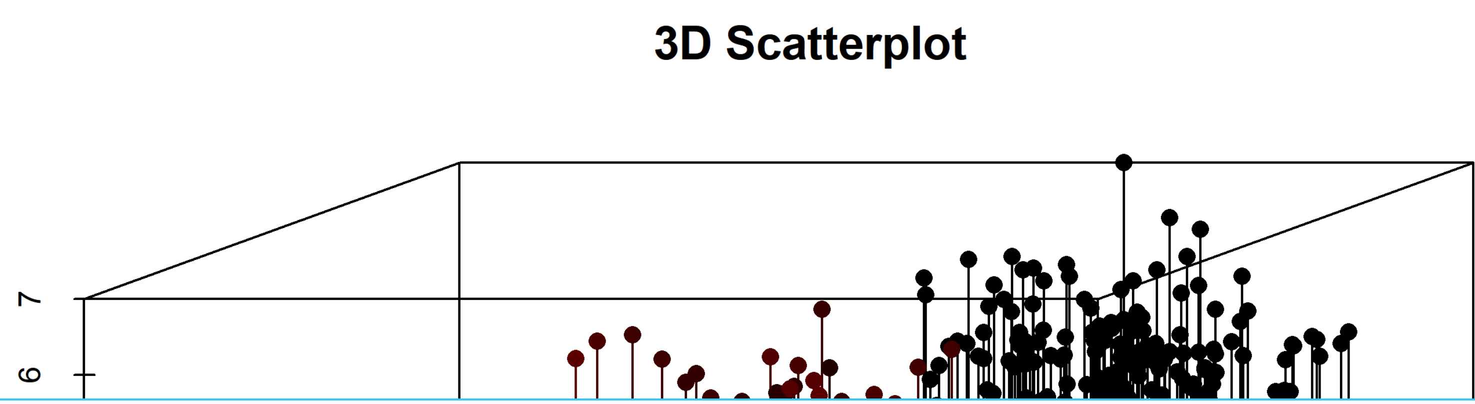
Strength Code

```
library(ggplot2)
library(gridExtra)

set.seed(123)  # for reproducibility

# Function to generate data with different strengths of relationship
generate_data <- function(n, strength) {
  x <- rnorm(n)
  y <- strength * x + rnorm(n, sd = sqrt(1 - strength^2))
  data.frame(x = x, y = y)
}

# Generate data
weak_data <- generate_data(100, 0.3)
moderate_data <- generate_data(100, 0.6)
strong_data <- generate_data(100, 0.9)

# Create plots
plot_data <- function(data, title) {
  ggplot(data, aes(x, y)) +
    geom_point(color = "blue", alpha = 0.6) +
    geom_smooth(method = "lm", se = FALSE, color = "red") +
    ggtitle(title) +
    theme_minimal() +
    theme(plot.title = element_text(hjust = 0.5))
}

weak_plot <- plot_data(weak_data, "Weak Relationship")
moderate_plot <- plot_data(moderate_data, "Moderate Relationship")
strong_plot <- plot_data(strong_data, "Strong Relationship")

# Combine plots
grid.arrange(weak_plot, moderate_plot, strong_plot, ncol = 3)
```



Simpsons's Code

```
library(scatterplot3d)
set.seed(123)
n = 1000
Education = rbinom(n, 2, 0.5)
Extraversion = rnorm(n) + Education
Salary = Education * 2 + rnorm(n) - Extraversion * 0.3
Salary = sample(10000:11000, 1) + rescale(Salary, to = c(0, 100000))
Extraversion = rescale(Extraversion, to = c(0, 7))
Education = factor(Education, labels = c("Low", "Medium", "High"))

data <- data.frame(Salary, Extraversion, Education)

attach(data)
s3d1 <- scatterplot3d(Salary,
  Education, Extraversion,
  pch=16, highlight.3d=TRUE,
  type="h", main="3D Scatterplot")
```



Anscombe's Code

```
cor1 <- format(cor(anscombe$x1, anscombe$y1), digits=4)
cor2 <- format(cor(anscombe$x2, anscombe$y2), digits=4)
cor3 <- format(cor(anscombe$x3, anscombe$y3), digits=4)
cor4 <- format(cor(anscombe$x4, anscombe$y4), digits=4)

#define the OLS regression
line1 <- lm(y1 ~ x1, data=anscombe)
line2 <- lm(y2 ~ x2, data=anscombe)
line3 <- lm(y3 ~ x3, data=anscombe)
line4 <- lm(y4 ~ x4, data=anscombe)

circle.size = 5
colors = list('red', '#0066CC', '#4BB14B', '#FCE638')

#plot1
plot1 <- ggplot(anscombe, aes(x=x1, y=y1)) + geom_point(size=circle.size, pch=21, fill=colors[[1]]) +
  geom_abline(intercept=line1$coefficients[1], slope=line1$coefficients[2]) +
  annotate("text", x = 12, y = 5, label = paste("correlation = ", cor1))

#plot2
plot2 <- ggplot(anscombe, aes(x=x2, y=y2)) + geom_point(size=circle.size, pch=21, fill=colors[[2]]) +
  geom_abline(intercept=line2$coefficients[1], slope=line2$coefficients[2]) +
  annotate("text", x = 12, y = 3, label = paste("correlation = ", cor2))

#plot3
plot3 <- ggplot(anscombe, aes(x=x3, y=y3)) + geom_point(size=circle.size, pch=21, fill=colors[[3]]) +
  geom_abline(intercept=line3$coefficients[1], slope=line3$coefficients[2]) +
  annotate("text", x = 12, y = 6, label = paste("correlation = ", cor3))

#plot4
plot4 <- ggplot(anscombe, aes(x=x4, y=y4)) + geom_point(size=circle.size, pch=21, fill=colors[[4]]) +
  geom_abline(intercept=line4$coefficients[1], slope=line4$coefficients[2]) +
  annotate("text", x = 15, y = 6, label = paste("correlation = ", cor4))

grid.arrange(plot1, plot2, plot3, plot4, top='Anscombe Quadrant -- Correlation Demostration')
```